Universiteit Leiden
The Netherlands

# Altered white matter microstructural organization in posttraumatic stress disorder across 3047 adults: results from the PGC-ENIGMA PTSD consortium

Dennis, E.L.; Disner, S.G.; Fani, N.; Salminen, L.E.; Logue, M.; Clarke, E.K.; ... ; Morey, R.A.

**Citation**
Dennis, E. L., Disner, S. G., Fani, N., Salminen, L. E., Logue, M., Clarke, E. K., … Morey, R. A. (2021). Altered white matter microstructural organization in posttraumatic stress disorder across 3047 adults: results from the PGC-ENIGMA PTSD consortium. *Molecular Psychiatry*, *26*(8), 4315-4330. doi:10.1038/s41380-019-0631-x

| | |
|---|---|
| Version: | Publisher's Version |
| License: | Creative Commons CC BY-SA 4.0 license |
| Downloaded from: | https://hdl.handle.net/1887/3276532 |

**Note:** To cite this publication please use the final published version (if applicable).

Molecular Psychiatry (2021) 26:4315–4330
https://doi.org/10.1038/s41380-019-0631-x

ARTICLE

# Altered white matter microstructural organization in posttraumatic stress disorder across 3047 adults: results from the PGC-ENIGMA PTSD consortium

Emily L. Dennis[1,2,3,4] et al.

Received: 20 June 2019 / Revised: 20 November 2019 / Accepted: 2 December 2019 / Published online: 19 December 2019
© The Author(s), under exclusive licence to Springer Nature Limited 2019

## Abstract

A growing number of studies have examined alterations in white matter organization in people with posttraumatic stress disorder (PTSD) using diffusion MRI (dMRI), but the results have been mixed which may be partially due to relatively small sample sizes among studies. Altered structural connectivity may be both a neurobiological vulnerability for, and a result of, PTSD. In an effort to find reliable effects, we present a multi-cohort analysis of dMRI metrics across 3047 individuals from 28 cohorts currently participating in the PGC-ENIGMA PTSD working group (a joint partnership between the Psychiatric Genomics Consortium and the Enhancing NeuroImaging Genetics through Meta-Analysis consortium). Comparing regional white matter metrics across the full brain in 1426 individuals with PTSD and 1621 controls (2174 males/873 females) between ages 18–83, 92% of whom were trauma-exposed, we report associations between PTSD and disrupted white matter organization measured by lower fractional anisotropy (FA) in the tapetum region of the corpus callosum (Cohen's $d = -0.11$, $p = 0.0055$). The tapetum connects the left and right hippocampus, for which structure and function have been consistently implicated in PTSD. Results were consistent even after accounting for the effects of multiple potentially confounding variables: childhood trauma exposure, comorbid depression, history of traumatic brain injury, current alcohol abuse or dependence, and current use of psychotropic medications. Our results show that PTSD may be associated with alterations in the broader hippocampal network.

## Introduction

Posttraumatic stress disorder (PTSD) is a debilitating mental health condition with a lifetime prevalence varying globally between 1 and 9% [1], with higher rates in women. Rates of PTSD are higher in populations exposed to greater levels of trauma, such as combat veterans [2] and civilians in conflict zones [3]. In addition to trauma type, genetics, and other sociological, psychological, and biological factors, individual differences in brain structure and function may explain vulnerability to developing PTSD following exposure to trauma, may result from trauma, or may be exacerbated by PTSD [4]. There is a lack of mechanistic evidence on the effects of stress and trauma on white matter structure. Exposure to trauma could lead to white matter damage, as excessive glucocorticoid levels can be neurotoxic and can impact myelination [5, 6]. Diffusion MRI (dMRI) is able to model white matter tracts and assess microstructural organization [7]. Fractional anisotropy (FA) is the most commonly used metric of microstructural organization, reflecting the degree to which water is diffusing along the axon (axially) as compared with across it (radially). Greater FA can reflect higher myelination, axonal diameter, or fiber density. Mean diffusivity (MD) reflects the average magnitude of diffusion across all directions, axial diffusivity (AD) is diffusion along the primary eigenvector (the dominant fiber direction), and radial diffusivity (RD) estimates diffusion perpendicular to the primary eigenvector. Altered microstructural organization is associated with several

These authors contributed equally: Neda Jahanshad, Rajendra A. Morey

**Supplementary information** The online version of this article (https://doi.org/10.1038/s41380-019-0631-x) contains supplementary material, which is available to authorized users.

✉ Emily L. Dennis
emily.dennis@hsc.utah.edu

Extended author information available on the last page of the article

different psychiatric disorders and could constitute a risk factor and/or a consequence of the disorders.

Studies of white matter microstructure in PTSD have reported inconsistent results. The majority report that PTSD is associated with lower FA [8–24], but some report higher FA [25–31], higher and lower FA in different regions [32], or null results [33–35]. Alterations in the cingulum bundle are frequently reported [9–13, 16, 18, 21, 23–29, 31, 32, 36], with differences also observed in the uncinate, corpus callosum, and *corona radiata* [14, 16, 18, 19, 24, 26, 29]. Inconsistent findings may be partially due to the use of hypothesis-driven rather than whole brain approaches, choice of analytic pipeline, selection of diffusion metrics, gender-specific studies, homogeneity of single cohort samples such as trauma-exposed vs. unexposed controls, and focus on military vs. civilian samples.

The PGC-ENIGMA PTSD working group is an international collaborative effort of the Psychiatric Genomics Consortium and the Enhancing NeuroImaging Genetics through Meta-Analysis (ENIGMA) consortium that aims to increase statistical power through meta- and mega-analyses of PTSD neuroimaging biomarkers. This collaborative approach has led to the largest PTSD neuroimaging study to date, reporting smaller hippocampal volume in patients with PTSD compared to controls [37]. Here we applied this approach to investigate the microstructural organization of white matter in PTSD. The ENIGMA Diffusion Tensor Imaging (DTI) workflow [38], which has successfully identified white matter compromise in schizophrenia [39], bipolar disorder [40], major depression [41], 22q11.2 deletion syndrome [42], and obsessive compulsive disorder [43], among others, was used in this study by 28 cohorts to process their dMRI data locally. We hypothesized that the largest effects of compromised microstructure would be evident in the fronto-limbic tracts, such as the cingulum, uncinate, fornix, mid-sagittal corpus callosum and tapetum; these tracts are strongly implicated in behavioral deficits of PTSD such as emotion regulation, working memory, and episodic memory [10, 16, 19, 20, 24, 44].

## Materials and methods

### Study samples

The PGC-ENIGMA PTSD dMRI analysis included 28 cohorts from seven countries totaling 1621 healthy controls and 1426 individuals with PTSD (either formally diagnosed or with CAPS-4 > 40, see Supplementary Fig. 1). The age range across cohorts was 18–83 years; all but two older Vietnam era cohorts had an average age between 29 and 50. Of the 3047 participants included in these analyses, 2071 (68%) were from military cohorts, which resulted in a disproportionate number of males (71%). The majority of cohorts included trauma-exposed controls (e.g., combat, community violence, intimate partner violence, $N = 1498$), although some included trauma-unexposed controls ($N = 123$), and one included no control group. Table 1 contains demographic and clinical information for each cohort. All participants provided written informed consent approved by local institutional review boards. Quality control was completed by each site, with visual quality checking and outlier detection. Details on ENIGMA-DTI methods [39], inclusion/exclusion criteria, and clinical information may be found in Supplementary Note 1, Supplementary Table 1, and Supplementary Note 2, respectively.

### Image acquisition and processing

The acquisition parameters for each cohort are provided in Supplementary Table 2. Preprocessing, including eddy current correction, echo-planar imaging-induced distortion correction and tensor fitting, was carried out at each site. Recommended protocols and quality control procedures are available on the ENIGMA-DTI and NITRC (Neuroimaging Informatics Tools and Resources Clearinghouse) webpages. Harmonization of preprocessing schemes was not enforced across sites to accommodate site- and acquisition-specific pipelines. Once tensors were estimated, they were mapped to the ENIGMA-DTI template and projected onto the ENIGMA-DTI template and were averaged within ROIs (http://enigma.ini.usc.edu/protocols/dti-protocols/). Further details and ROI abbreviations can be seen in Supplementary Note 1. In a subset of sites we extracted motion parameters from the eddy current correction step to determine if motion played a role in our case control findings. We examined rotation and translation averaged across the $X$, $Y$, and $Z$ axes and found a marginally significant difference between PTSD and control groups in average rotation ($d = 0.12$ $p = 0.030$). Follow-up analyses including motion parameters as covariates yielded results that were consistent with our main analyses (for details see Supplementary Note 1 and Supplementary Fig. 2).

### Statistical analysis

For each cohort/study, a linear model was fit using the *ppcor* and *matrixStats* packages in R 3.1.3, with the ROI FA as the response variable and PTSD and covariates as predictors. For cohorts/studies including more than one data collection site, site was included as a fixed dummy variable in the site-level analysis. As in prior ENIGMA disease working group meta-analyses [39], a random-

**Table 1** Demographic information on adult cohorts included in analyses.

| Site | Total *N* | M | F | *N* PTSD | *N* Control | Age range | Average age | PTSD scale | Depression scale | Type of controls | Dataset |
|---|---|---|---|---|---|---|---|---|---|---|---|
| ADNI-DoD | 134 | 134 | 0 | 70 | 64 | 61–83 | 69.3 | CAPS-4 | GDS | Exposed | Military |
| Beijing | 67 | 32 | 35 | 32 | 35 | 37–61 | 49.4 | PCL-5 | na | Exposed | Civilian |
| Booster | 70 | 36 | 34 | 34 | 36 | 22–59 | 39.9 | CAPS-4 | HADS-D | Exposed | Police |
| Columbia | 33 | 24 | 9 | 16 | 17 | 20–58 | 36.1 | CAPS-4 | HAM-D | Exposed | Civilian |
| Duke-1 | 187 | 142 | 45 | 50 | 137 | 21–57 | 39.4 | CAPS-4/5 | BDI | Exposed | Military |
| Duke-2 | 88 | 61 | 27 | 19 | 69 | 23–66 | 39.9 | SCID/DTS | na | Exposed | Military |
| Duke-3 | 61 | 50 | 11 | 18 | 43 | 23–65 | 38.8 | CAPS-4/5 | na | Exposed | Military |
| Grady trauma project | 132 | 0 | 132 | 50 | 82 | 18–62 | 39.6 | CAPS-4 | BDI | Exposed | Civilian |
| Groningen | 49 | 0 | 49 | 49 | 0 | 23–58 | 40.3 | CAPS-4 | BDI | No controls | Civilian |
| INTRuST | 214 | 117 | 97 | 77 | 137 | 18–56 | 36 | MINI/CAPS-4 /PCL-M/SCID | na | Exposed | Military and civilian |
| iSCORE | 99 | 86 | 13 | 44 | 55 | 19–51 | 35.8 | PCL-M | CES-D | Exposed | Military |
| Lawson | 98 | 52 | 46 | 46 | 52 | 18–59 | 34.7 | CAPS-4 | BDI | Exposed and unexposed | Civilian |
| McLean | 55 | 0 | 55 | 41 | 14 | 18–62 | 37 | CAPS-5 | BDI | Exposed | Civilian |
| Münster | 25 | 0 | 25 | 14 | 11 | 19–51 | 29 | SCID | BDI | Exposed | Civilian |
| New South Wales | 162 | 62 | 100 | 85 | 77 | 18–69 | 40.2 | CAPS-4 | HAM-D | Exposed | Civilian |
| South Dakota | 90 | 81 | 9 | 55 | 35 | 22–45 | 31.8 | PCL-M | na | Exposed | Military |
| Stellenbosch-1 | 71 | 20 | 51 | 27 | 44 | 21–77 | 48.0 | CAPS-5 | na | Both | Civilian |
| Stellenbosch-2 | 31 | 19 | 12 | 17 | 14 | 21–66 | 36.6 | CAPS-5 | na | Both | Civilian |
| U Sydney | 64 | 34 | 30 | 31 | 33 | 17–49 | 36.25 | CAPS-4 | DASS | Exposed | Civilian |
| UMC Utrecht | 94 | 94 | 0 | 46 | 48 | 21–57 | 35.6 | CAPS-4 | SCID | Both | Military |
| UW-Madison | 48 | 44 | 4 | 25 | 23 | 22–48 | 31 | CAPS-4 | BDI | Exposed | Military |
| VA Boston | 493 | 456 | 37 | 305 | 188 | 18–65 | 31.2 | CAPS-4 | na | Exposed | Military |
| VA Houston | 69 | 44 | 25 | 53 | 16 | 21–58 | 31.4 | CAPS-4 | BDI | Exposed | Military |
| VA Minneapolis-1 | 124 | 120 | 4 | 49 | 75 | 23–62 | 34.2 | CAPS-4 | SCID | Both | Military |
| VA Minneapolis-2 | 130 | 121 | 9 | 67 | 63 | 22–59 | 32.9 | CAPS-4 | SCID | Both | Military |
| VA Waco | 53 | 46 | 7 | 36 | 17 | 25–60 | 39.6 | PCL-5 | na | Unexposed | Military |
| VETSA | 239 | 239 | 0 | 33 | 206 | 56–66 | 61.8 | PCL-C | CES-D | Exposed | Military |
| Yale/NCPTSD | 67 | 60 | 7 | 37 | 30 | 21–60 | 34.1 | CAPS-4 | BDI | Exposed | Military |
| Overall | **3047** | **2174** | **873** | **1426** | **1621** | | **39.6** | | | | |

effects inverse-variance weighted meta-analysis was conducted at a central coordinating site (the University of Southern California Imaging Genetics Center) in R (metafor package, version 1.99–118 http://www.metafor-project.org/) to combine individual cohort estimated effect sizes. Cohen's *d* for the main effect of group and unstandardized β coefficients (regression parameters) for continuous predictors were computed with 95% confidence intervals. We used the Cohen's d calculation that accounts for covariates in the fixed effects model, using the following equation:

$$d = \mathrm{M1} - \mathrm{M2}/\text{pooled SD, where pooled}$$
$$\mathrm{SD} = \sqrt{((\mathrm{SD}_1^2 + \mathrm{SD}_2^2)/2)}$$

Heterogeneity scores ($I^2$) for each test were computed, indicating the percent of total variance in effect size explained by heterogeneity across cohorts. Bilaterally averaged FA was the primary imaging measure, with corresponding MD, RD, and AD examined post hoc when FA was significant for an effect of diagnosis. Lateralized ROIs were examined post hoc when a significant association was found with the bilateral average. The corticospinal tract was not analyzed as it has poor reliability [38]. The average correlation in FA between all ROIs was $r = 0.61$. A Bonferroni correction is overly conservative when there are correlations between the multiple dependent measures being tested. Therefore, we calculated the effective number of independent tests based on the observed correlation structure between the alternate responses using two different methods. The equation proposed by Nyholt [45] yielded an estimated 13.3529 effective tests (Meff), giving a significance threshold of $p < 0.05/13.3529 = 0.0037$. The equation of Li and Ji [46] yielded Veff = 9, giving a significance threshold of

$p < 0.05/9 = 0.0057$. Recent empirical work based on simulations concludes that when the number of effective tests were known, the Nyholt calculation is less accurate than the method of Li and Ji because it overestimates the number of effective variables [47]. Given the lack of consensus in the field along with initial evidence that the Li and Ji threshold is more accurate, we adopted this method but also note whether results survived the Nyholt threshold.

### Nonlinear age term

We first conducted analyses to examine whether a nonlinear age term should be included in statistical models along with age and sex, as age has a nonlinear effect on FA [48]. As this analysis did reveal a significant effect of nonlinear age above and beyond linear age, age$^2$ was included in all subsequent analyses. Demographic details are shown in Supplementary Fig. 3.

### Primary group comparison

We compared PTSD cases to all controls (both trauma-exposed and unexposed), PTSD cases to trauma-exposed controls only, and trauma-exposed to trauma unexposed controls. Given the significant impact of trauma exposure on brain structure [49], we compared trauma-exposed with unexposed controls to examine whether potential differences may be attributed to trauma exposure rather than PTSD. We repeated analyses separately in sites acquiring 30 or fewer diffusion directions, sites acquiring more than 30 directions, sites using isotropic voxel dimensions, and sites using anisotropic voxel dimensions. These are discussed in further detail in Supplementary Note 3, Supplementary Table 3, and Supplementary Fig. 4. Secondary subgroups: We examined PTSD associations in males and females separately, and in military and civilian samples separately. These results may be found in Supplementary Note 4. Secondary interactions: We examined potential interactions between PTSD and age or sex. These results can be seen in Supplementary Note 5.

### Secondary additional covariates

We tested a model including ancestry, but as this was a meta-analysis and most cohorts were primarily composed of participants of white non-Hispanic descent, this had a very minimal impact, and we did not include this variable as a covariate in our analysis. We examined the impact of five potentially confounding covariates on the associations of PTSD with FA—childhood trauma, depression, alcohol dependence/abuse, traumatic brain injury (TBI, of any severity), and use of psychotropic medications. We compared the white matter microstructure of individuals with PTSD to that of controls with each covariate included individually in the model, and in the subset of sites that collected data on childhood trauma, depression, alcohol use disorders, TBI, or medication without that covariate in the model to determine whether differences in results were due to the inclusion of the covariate or the reduction in sample size. There were not enough participants with all five variables to simultaneously model these potential confounds in a single model. Details of these methods and results are provided in Supplementary Note 6. Briefly, binary variables were created for depression, TBI, and medication use. As depression was assessed using a variety of measures, we used published cut-offs to recode the data as categorical depression (see Supplementary Note 2 for more details). Alcohol use disorders and childhood trauma were coded as three-level ordinal variables based on evidence of dose-dependent effects on brain structure and clinical severity, respectively [50, 51]: Alcohol use disorders: 0 = no alcohol use disorder, 1 = alcohol abuse, 2 = alcohol dependence, as measured by the SCID or AUDIT [52]; childhood trauma (as measured by the Childhood Trauma Questionnaire): 0 = no reported childhood trauma, 1 = one type of childhood trauma exposure, 2 = two or more types of childhood trauma exposure; *PTSD severity*: To examine PTSD severity, we conducted linear regressions on CAPS-4 score in the PTSD group for sites that collected CAPS-4. We examined linear associations with CAPS-4 score covarying for childhood trauma, depression, alcohol use disorders, TBI, and medication use, and we tested associations with CAPS-4 separately in military veterans and civilians as well as males and females (see Supplementary Note 6 for more details).

## Results

### Group differences

We found significantly lower FA in the PTSD group in the tapetum of the corpus callosum ($d = -0.11$, $p = 0.0055$) when comparing PTSD ($n = 1{,}377$) and all controls ($n = 1{,}620$). This result did not survive the more conservative but potentially less accurate Nyholt threshold. Post hoc analysis revealed a larger effect in the left than in the right tapetum (left $d = -0.13$, $p = 0.00090$; right $d = -0.067$, $p = 0.042$). Post hoc analysis also revealed higher RD in the tapetum in the PTSD group (bilateral $d = 0.09$, $p = 0.027$; left $d = 0.11$, $p = 0.0038$).

In the analysis comparing participants with PTSD ($n = 1{,}319$) to trauma-exposed controls ($n = 1{,}498$), we found

**Table 2** Results from the group comparisons.

| | PTSD vs all controls[a] | | | | PTSD vs trauma-exposed controls[b] | | | |
|---|---|---|---|---|---|---|---|---|
| ROI | Meta *d* | Meta *p* value uncorrected | 95% CI | $I^2$ | Meta *d* | Meta *p* value uncorrected | 95% CI | $I^2$ |
| Average FA | −0.02 | 0.69 | [−0.09, 0.06] | 0 | −0.03 | 0.48 | [−0.11, 0.05] | 0 |
| ACR | −0.01 | 0.85 | [−0.09, 0.07] | 7.51 | −0.03 | 0.48 | [−0.11, 0.05] | 0 |
| ALIC | −0.04 | 0.26 | [−0.12, 0.03] | 0 | −0.04 | 0.38 | [−0.11, 0.04] | 0 |
| BCC | −0.04 | 0.30 | [−0.12, 0.04] | 0 | −0.05 | 0.21 | [−0.13, 0.03] | 0 |
| CC | −0.05 | 0.24 | [−0.12, 0.03] | 0 | −0.05 | 0.25 | [−0.12, 0.03] | 0 |
| CGC | −0.03 | 0.43 | [−0.11, 0.05] | 0 | −0.03 | 0.42 | [−0.11, 0.05] | 0 |
| CGH | 0.02 | 0.67 | [−0.06, 0.09] | 0 | 0.01 | 0.72 | [−0.06, 0.09] | 0.01 |
| CR | −0.02 | 0.52 | [−0.10, 0.05] | 0 | −0.04 | 0.37 | [−0.11, 0.04] | 0 |
| EC | 0.03 | 0.37 | [−0.04, 0.11] | 0 | 0.03 | 0.52 | [−0.05, 0.10] | 0 |
| FX | −0.02 | 0.67 | [−0.10, 0.07] | 14.33 | −0.02 | 0.61 | [−0.12, 0.07] | 20.05 |
| FXST | 0.00 | 0.95 | [−0.08, 0.07] | 0 | 0.00 | 0.96 | [−0.08, 0.08] | 0.01 |
| GCC | −0.01 | 0.80 | [−0.09, 0.07] | 0 | −0.01 | 0.74 | [−0.09, 0.07] | 0 |
| IC | 0.00 | 0.97 | [−0.07, 0.08] | 0.01 | 0.01 | 0.84 | [−0.07, 0.09] | 0 |
| PCR | −0.04 | 0.25 | [-0.12, 0.03] | 0 | −0.04 | 0.37 | [−0.11, 0.04] | 0 |
| PLIC | 0.03 | 0.51 | [−0.06, 0.13] | 27.71 | 0.04 | 0.39 | [−0.05, 0.14] | 24.80 |
| PTR | −0.03 | 0.53 | [−0.11, 0.06] | 10.68 | −0.03 | 0.48 | [−0.12, 0.06] | 11.95 |
| RLIC | 0.00 | 1.0 | [−0.08, 0.08] | 0 | 0.00 | 0.97 | [−0.08, 0.08] | 0 |
| SCC | −0.08 | 0.096 | [−0.18, 0.01] | 31.36 | −0.06 | 0.23 | [−0.17, 0.04] | 33.81 |
| SCR | −0.02 | 0.60 | [−0.10, 0.06] | 0 | −0.03 | 0.51 | [−0.11, 0.05] | 0 |
| SFO | *−0.10* | *0.014* | *[−0.17,−0.02]* | *0* | *−0.11* | *0.0073* | *[−0.19, −0.03]* | *0* |
| SLF | 0.03 | 0.49 | [−0.05, 0.10] | 0 | 0.04 | 0.35 | [−0.04, 0.12] | 0 |
| SS | −0.02 | 0.64 | [−0.09, 0.06] | 0 | −0.03 | 0.39 | [−0.11, 0.04] | 0 |
| TAP | **−0.11** | **0.0055** | **[−0.18, −0.03]** | **0** | *−0.10* | *0.011* | *[−0.18, −0.02]* | *0* |
| UNC | 0.02 | 0.58 | [−0.06, 0.10] | 0.01 | 0.01 | 0.76 | [−0.07, 0.09] | 0 |

Cohen's *d* values, uncorrected *p* values, the 95% confidence interval for the *d* statistic, and the $I^2$ (heterogeneity) are shown for the group comparisons. **Bolded** results are significant when corrected for multiple comparisons, *italicized* results are marginally significant (based on the Li and Ji adjusted Bonferroni correction)

[a]Results comparing PTSD to all controls

[b]Results comparing PTSD to trauma-exposed controls only

lower FA in the left tapetum in PTSD ($d = -0.13$, $p = 0.0014$), although the effect in bilateral tapetum did not survive correction for multiple comparisons but had comparable effect size ($d = -0.10$, $p = 0.011$) (see Table 2 and Fig. 1, see Fig. 2 for site-specific effects, and Fig. 3 for a neuroanatomical figure). PTSD participants from cohorts that only included trauma-unexposed controls were not included.

Comparing trauma-exposed ($n = 200$) to trauma unexposed controls ($n = 93$) from six sites, we found an effect that did not withstand correction for multiple comparisons but had a comparable effect size of lower FA in exposed controls in the tapetum, splenium of corpus callosum, and fornix/stria-terminalis ($d = -0.41$, $p = 0.014$; $d = -0.48$, $p = 0.0042$; $d = -0.36$, $p = 0.019$, respectively), along with significantly higher MD and RD in the splenium ($d = 0.48$, $p = 0.0017$; $d = 0.56$, $p = 0.00023$, respectively) and an effect that did not survive correction for multiple comparisons but had a comparable effect size of higher RD in the tapetum ($d = 0.32$, $p = 0.036$). Lastly, for a subset of cohorts we had data on lifetime PTSD (444 lifetime PTSD vs. 230 controls across seven cohorts). We did not find any significant results for this comparison.

## Subgroups

We examined military vs. civilian cohorts, and male vs. female participants separately. We found significantly lower FA in males in the superior fronto-occipital fasciculus (SFO), along with associations that did not survive correction for multiple comparisons with tapetum FA in the military-only and male-only subgroups separately (see Supplementary Note 4 and Supplementary

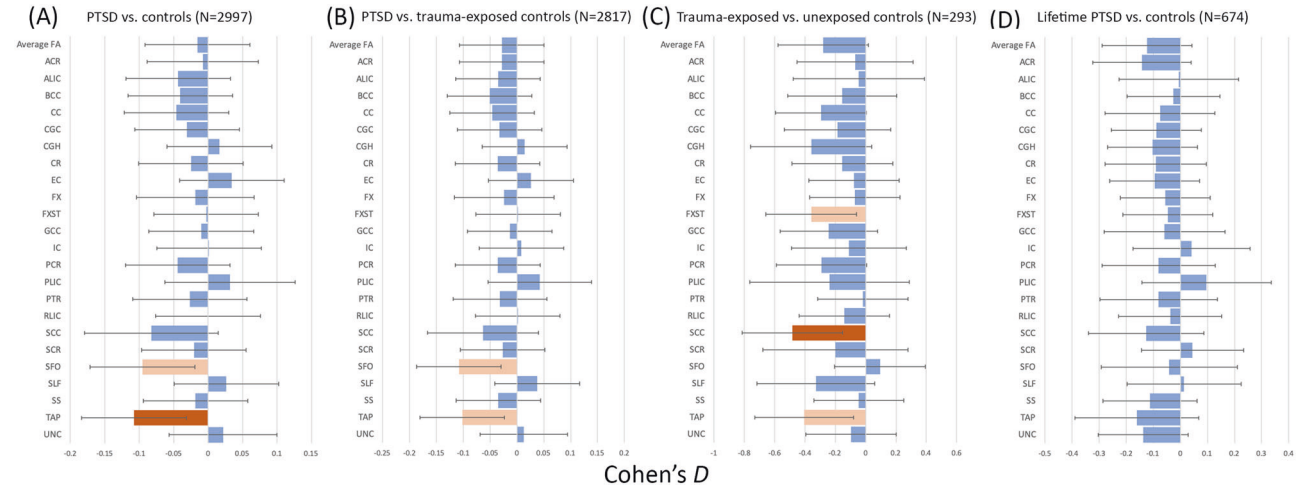



**Fig. 1 Results from the group comparisons. a** Results comparing PTSD to all controls; **b** results comparing PTSD to trauma-exposed controls only; **c** results comparing trauma-exposed to unexposed participants; **d** lifetime PTSD to controls. Cohen's *d* statistics are shown across all bilateral and midline ROIs and average FA, with bars indicating the 95% confidence interval. The ROI abbreviations are explained in Supplementary Note 1. As PTSD was coded "1" and control "0", negative statistics indicate lower FA in PTSD. Total *N* is listed for each comparison. Dark orange bars indicate significance ($p < 0.0057$) and light orange bars indicate results that did not withstand correction for multiple comparisons ($0.05 > p > 0.0057$). Error bars are 95% CI.

Fig. 5 for more details). Results of group-by-sex and group-by-age interactions were not significant and are shown in Supplementary Note 5 and Supplementary Fig. 6.

## Additional covariates

The role of potentially confounding variables on the association between PTSD and the tapetum was tested in several post hoc analyses focused on left, right, and bilateral tapetum FA (see Supplementary Fig. 7). As these analyses were considered post hoc and limited to the tapetum, we used a test-wise significance threshold of $p < 0.05$. Results generally remained significant across all models. *Depression*: Including dichotomous depression as a covariate (699 PTSD vs. 864 controls) resulted in lower bilateral and lower left tapetum FA in the PTSD group (bilateral $d = -0.12$, $p = 0.036$; left $d = -0.16$, $p = 0.0024$). *Alcohol Use Disorders*: Including AUD as a covariate (633 PTSD vs. 524 controls) resulted in lower bilateral and left tapetum FA in the PTSD group (bilateral $d = -0.14$, $p = 0.018$; left $d = -0.16$, $p = 0.0081$) and borderline lower right tapetum FA ($d = -0.11$, $p = 0.077$). *Traumatic Brain Injury*: Including a binary TBI variable (830 PTSD vs. 1035 controls) resulted in lower left tapetum FA ($d = -0.11$, $p = 0.023$). *Medication Use*: Including a dichotomous psychotropic medication covariate (694 PTSD vs. 663 controls) resulted in lower left tapetum FA in the PTSD group (left $d = -0.012$, $p = 0.028$). *Childhood Trauma*: Including childhood trauma as a covariate (367 PTSD vs. 598 controls) did not yield any significant results, but neither did the analysis in the reduced sample without CT in the model, suggesting that the sample reduction impacted these results. To control for covariate- and cohort-dependent changes in sample size, each analysis was repeated in a smaller sample that corresponded to omitting the relevant covariate. The tapetum results remained consistent in nearly all reduced sample analyses—significant effects survived covariate adjustment and effects that disappeared (such as with childhood trauma) were also absent in the reduced sample. Thus, covariates had minimal impact beyond the reduction in sample size (see Supplementary Note 6 and Supplementary Figs. 8–12). A table showing how many participants at each site had information on these potentially confounding variables may be found in Supplementary Table 4.

## PTSD severity

Examining PTSD symptom severity in subjects assessed with CAPS-4 ($N = 1{,}764$ from 17 sites), we found an association, which did not survive correction for multiple comparisons, between higher CAPS-4 score and lower FA in the tapetum and SFO (tapetum: $\beta = -9.0 \times 10^{-5}$, $p = 0.024$; SFO: $\beta = -6.6 \times 10^{-5}$, $p = 0.028$; Fig. 4). PTSD symptom severity in the PTSD group (measured by the CAPS-4, $N = 956$ from 17 sites) was not associated with FA (Fig. 4). Subgroup analyses yielded associations that did not survive correction for multiple comparisons between CAPS-4 score and tapetum FA in the military cohorts and between CAPS-4 score and SFO FA in the

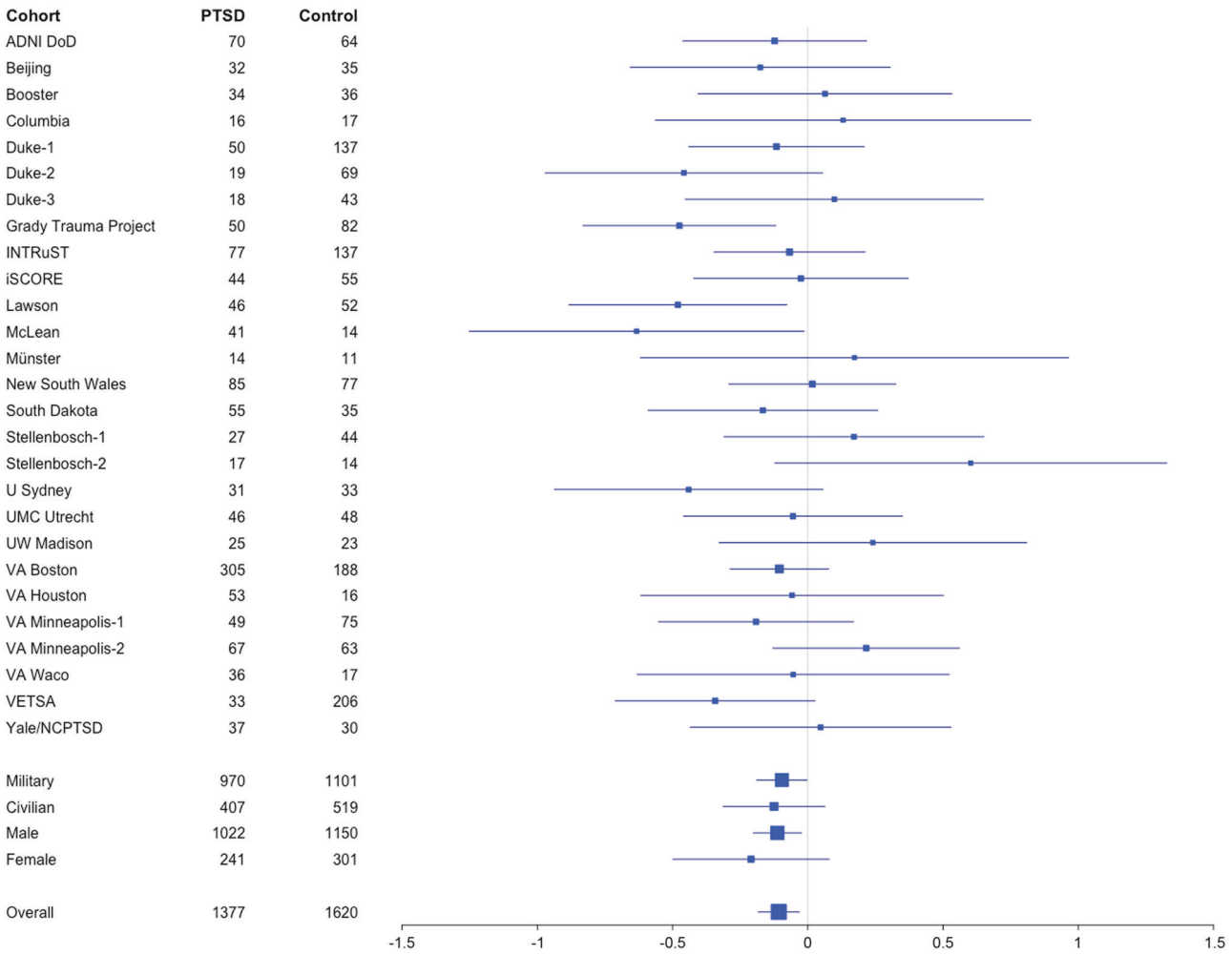



**Fig. 2 Site effects for tapetum result.** Forest plot shows the effect sizes (Cohen's *d*) for each of the 25 cohorts, scaled by sample size, with bars for 95% CI. The effect size and 95% CI of the meta-analysis is shown at the bottom of the figure, along with effect sizes and 95% CI for the subgroup analyses of the military cohorts, civilian cohorts, male cohorts, and female cohorts.

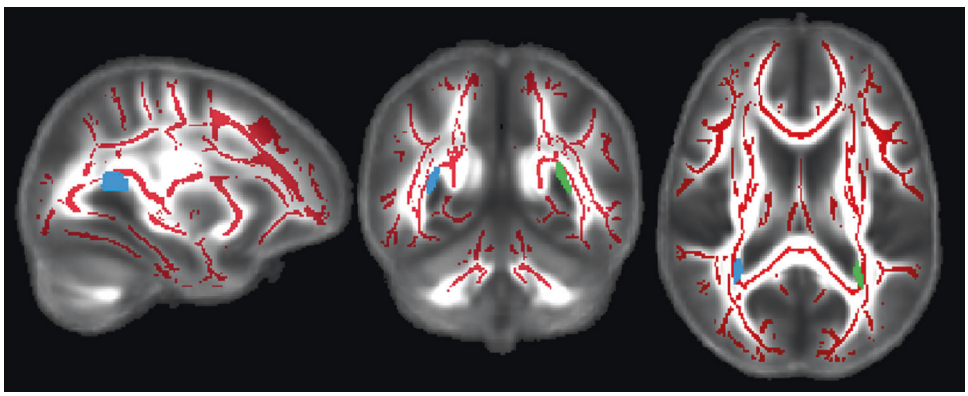


**Fig. 3 Tapetum displayed on the ENIGMA template FA.** The skeleton is shown in red, the left tapetum (green) and right tapetum (blue) ROIs are displayed. Left in image is right in brain.

male subgroup. Results were similar when potentially confounding variables were included, with no significant associations, although there was an effect in tapetum that did not withstand correction for multiple comparisons when psychotropic medication use was included. Detailed analyses of PTSD severity and covariates within subgroups are in Supplementary Note 6 and Supplementary Figs. 13 and 14.

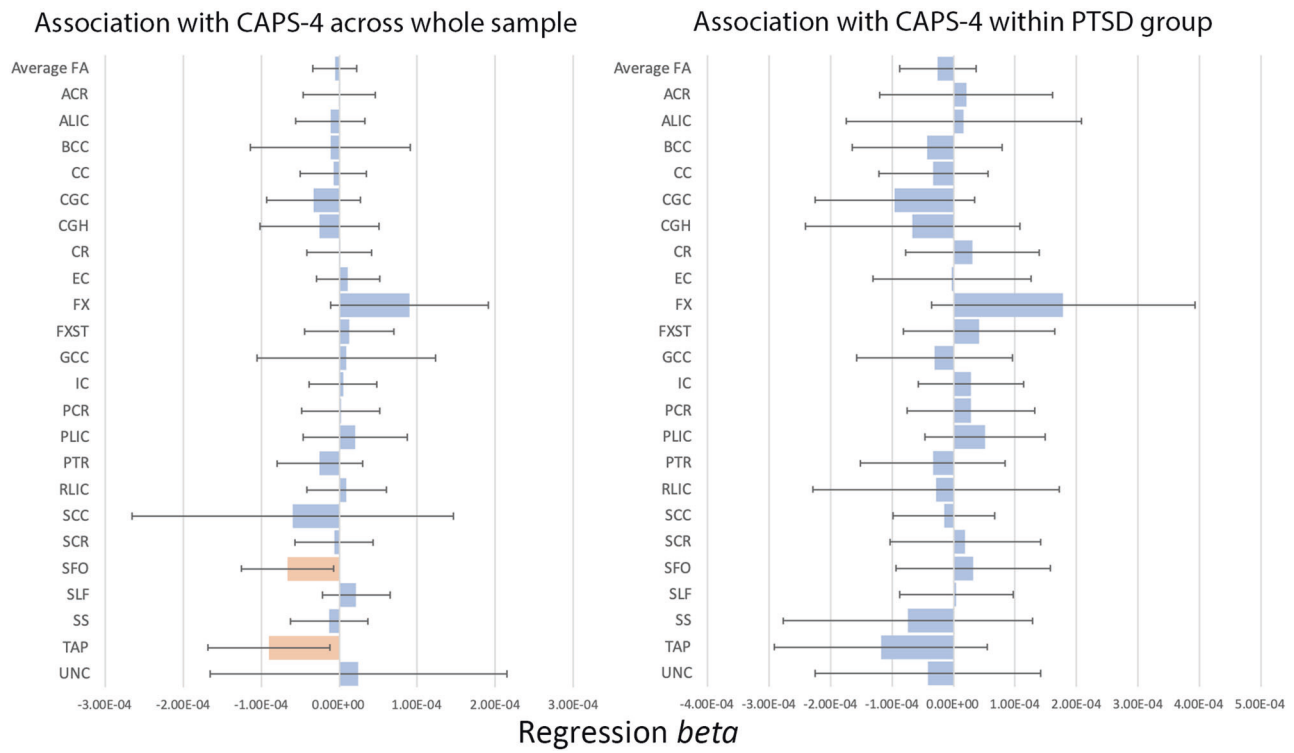



**Fig. 4 Linear association with CAPS-4 across the whole sample (left) and within the PTSD cases only (right).** Meta-regression unstandardized β statistics are shown across all bilateral and midline ROIs and average FA, with bars indicating the 95% confidence interval. The ROI abbreviations are explained in Supplementary Note 1. Light orange bars indicate results that did not withstand correction for multiple comparisons ($0.05 > p > 0.0057$). Error bars are 95% CI.

## Discussion

We present dMRI results from a multicohort study conducted by the PGC-ENIGMA PTSD consortium. In a meta-analysis of 3047 participants from 28 sites, we found lower FA and higher RD in the tapetum among adults with PTSD (neuroanatomical figure—Fig. 3), which remained after accounting for several potentially confounding factors. The tapetum is a major tract within the corpus callosum that serves as a conduit between right and left hippocampus. Prior studies of white matter disruption in PTSD have found alterations in other hippocampal tracts, but were generally hindered by small sample sizes leading to inconsistent findings across studies. Our results add to the existing literature in identifying structural disruptions that compromise putative hippocampal functions, which are known to play a central role in PTSD symptomatology [53, 54].

The tapetum is a small segment of the corpus callosum that connects the temporal lobes, in particular the left and right hippocampus [55]. It is one of the last corpus callosum segments to develop and experiences rapid growth around age 14, which may make it vulnerable to the effects of trauma for a longer period of time [56]. Structural and functional alterations in the hippocampus are frequently reported in PTSD, with smaller volumes [37], decreased activation, and disrupted functional connectivity with the medial and lateral prefrontal cortices [57, 58]. Disrupted functional connectivity between the left and right hippocampus has also been previously reported in PTSD [59]. Here we report microstructural evidence that *structural* connectivity between the left and right hippocampus may also be disrupted in PTSD.

We did not find PTSD-related effects in the cingulum, uncinate, fornix, and corpus callosum, as some prior studies of smaller, more homogeneous populations have shown [9–14, 16, 18–21, 23–29, 31, 32, 36, 60]. This may reflect a limitation in the meta-analysis of heterogeneous study samples, but this is an unlikely explanation given our low estimates of heterogeneity ($I^2$) in these tracts (<10%). However, these effects were not entirely absent in our subgroup analyses, suggesting that they may be more specific to population, trauma type, etc. While many studies have reported that PTSD is associated with alterations in the cingulum bundle [9–13, 16, 18, 21, 23–29, 31, 32, 36], which has a hippocampal component, the tapetum has not yet emerged for several possible reasons. Many prior studies took an ROI approach, which limited analyses to predetermined regions that frequently omitted the tapetum, a small region often grouped with other tracts such as the splenium or posterior thalamic radiation. Critically, in 2013, an error was uncovered in the JHU atlas used as part of the

TBSS pipeline, with the uncinate incorrectly identified as the inferior fronto-occipital fasciculus, and the tapetum incorrectly identified as the uncinate [61]. Thus, the tapetum was simply not examined in prior studies, with one very recent exception showing that tapetum abnormalities are associated with lower major depressive disorder remission [62]. There are further data suggesting that the absence of tapetum findings may partly relate to methodological limitations of prior studies; in a recent dMRI meta-analysis of attention deficit hyperactivity disorder (ADHD), lower tapetum FA was found to characterize ADHD, despite the fact that this specific region of the CC has not frequently emerged in ADHD studies [63]. Finally, the precise role of the tapetum in connecting the left and right hippocampus was only recently elucidated by mapping the subcortical connectome with exquisitely high-resolution mapping capable of discerning the intermingling of tapetum and other corpus callosum fibers [55].

Childhood trauma is the greatest single risk factor for future vulnerability to PTSD [64]; numerous studies show significant alterations in brain structure and function in individuals who experience significant early life stress [51, 65]. Some of these alterations likely contribute to a higher risk for psychopathology, but childhood trauma exposure alone did not explain the association between tapetum white matter disruption and PTSD that we report here. While it is likely that childhood trauma exposure plays some role in tapetum organization, these analyses indicate that our PTSD results are not solely due to confounding effects between PTSD and exposure to childhood trauma. Depression is frequently comorbid with PTSD [66] and is associated with disrupted white matter organization, although the affected tracts are broadly distributed [67]. Accounting for depression in group comparisons did not significantly alter our results, suggesting that tapetum white matter disruption is not confounded by depression. The UK Biobank sample revealed FA reductions associated with depression in commissural and association fibers, thalamic radiations, left superior longitudinal fasciculus (SLF), superior and forceps major [68]. The ENIGMA MDD Working Group reports alterations in the corpus callosum and hippocampal cingulum but not in the tapetum (labeled in that paper as the uncinate, $p = 0.172$) [69]. Particularly in military populations, which formed the majority of our sample, PTSD is often comorbid with traumatic brain injury (TBI) [70]. White matter is particularly vulnerable to TBI, which produces stretching and shearing of axons and altered neurometabolism [71]. Accounting for TBI also had some effect on our results but the left tapetum remained significant in post hoc analyses, indicating that TBI is not specifically associated with white matter damage in the tapetum despite numerous reports mild TBI is associated with white matter damage generally. Preliminary results from the ENIGMA Military Brain Injury group have not shown significant differences in the tapetum [72]. Psychotropic medications are another potential confound, given their neurotrophic and neuroprotective effects [73]. The result in the tapetum persisted after covarying for psychotropic medication, indicating that our findings are unlikely to be explained by medication. Lastly, PTSD can be comorbid with alcohol use disorders, which have a poorer clinical prognosis [74, 75]. Alcohol use disorders have been associated with significant changes in white matter organization [76, 77] but did not influence the present results. Although none of these potentially confounding variables fully explained our reported results, it is possible that altered tapetum microstructure could represent a more general response to trauma or stress. As shown in Fig. 1, altered tapetum FA is present in trauma-exposed controls compared to unexposed controls, although the sample size for this comparison was around 10% of the total sample size. The post hoc analysis of the tapetum was still significant comparing PTSD to trauma-exposed controls, suggesting a dose response, with some alterations related to trauma that does not lead to psychopathology but more in PTSD. Trauma is one form of severe stress, but we do not know how more minor stress may affect tapetum organization. Smaller hippocampal volume has been reported across disorders, suggesting that it may be a nonspecific marker of disease [78].

We found an association that did not withstand correction for multiple comparisons of PTSD with FA in the tapetum in male and military subgroups separately. Although results were nonsignificant in female or civilian subgroups, the effect size was slightly larger and in the same direction. The female and civilian subgroups were smaller and therefore the analyses had lower power than in male and military subgroups. Most prior dMRI studies in civilians report lower FA in PTSD [9, 10, 12–14, 16, 17, 21–25, 30–32, 60]. Studies of military cohorts have been mixed, reporting higher FA [26–29, 36], lower FA [8, 15, 18–20], and null results [33–35]. This discrepancy may be due to differences in age, chronicity, and type of trauma exposure, although military personnel may also experience civilian trauma. Combat-related PTSD is often comorbid with TBI, which is also associated with white matter disruption, constituting a potentially confounding factor for studies [72].

In the absence of longitudinal data, our analysis cannot make causal inferences nor make conclusions about the direction of the relationship between PTSD and tapetum white matter organization. Disrupted white matter of the tapetum may represent a vulnerability that predates the onset of PTSD, or a pathological response to trauma. In twins discordant for exposure to combat stress, the unexposed twins of combat veterans with PTSD have smaller

hippocampal volume than the unexposed twins of combat veterans without PTSD [79]. Individuals with two risk alleles of the *FKBP5* gene have demonstrated lower cingulum FA above and beyond the association of cingulum FA with PTSD [11, 60]. These studies suggest that heritable differences in brain structure may influence risk of developing PTSD. Evidence that alterations are caused by PTSD was observed in Israeli Defense Force recruits with reduced structural connectivity between the hippocampus and ventromedial prefrontal cortex, but only after exposure to military stress [80]. With the varying developmental trajectories of brain structure, function, and connectivity, along with the varying distribution of stress hormone receptors in the brain, the complex question of vulnerability vs. consequence will require prospective longitudinal neuroimaging studies.

Some evidence indicates that high FA is a marker of resilience to the effects of stress [81, 82]. A putative marker of resilience is the ability to attenuate stress-induced increases in corticotropin-releasing hormone and glucocorticoids through an elaborate negative feedback system, and to modulate the expression of brain-derived neurotrophic factor (BDNF) [83, 84]. BDNF has myriad functions including supporting neuronal differentiation, maturation, and survival [85, 86]. In particular, hippocampal BDNF is implicated in the development of neural circuits that promote stress adaptations [83]. These stress adaptation circuits involve white matter in the fornix and other fronto-limbic connections [87].

### Limitations

Our study has several limitations. One limitation of TBSS studies is the inability to fully attribute results to particular fiber bundles, especially given that the tapetum is a relatively small structure. Future studies may benefit by using tractography to more reliably identify the affected bundles, but this is difficult across sites. Second, not all participants classified as PTSD received clinician-administered interview (such as the CAPS) to confirm diagnoses. Third, we could not reliably measure chronicity across different cohorts. Other variables that we could not examine given the heterogeneity across sites include treatment effects, symptom clusters, and trauma types, as opposed to current PTSD diagnosis. Given that effects have not been previously reported in the tapetum, it would be helpful to know if altered tapetum organization is linked to particular symptoms or trauma type and we encourage researchers to examine this possibility in their datasets. Although we analyzed data from over 3000 participants, we may have been underpowered to examine group-by-sex interactions, as 55% of our sample came from cohorts including only males or only females or samples that were >90% male. Diffusion metrics are not scanner invariant, and can vary even in scanners of the same model. For this reason, we were limited to a meta-analytic approach which may have lower power than mega-analysis. However, studies including both meta- and mega-analyses of brain volume in other ENIGMA groups have found minimal differences [88, 89]. Lastly, our cross-sectional design did not allow us to investigate to what degree WM differences predate the onset of PTSD.

Future studies should further investigate the tapetum using high spatial and angular resolution tractography to replicate our findings. Future and existing studies with more in-depth phenotyping than was possible here could examine how alterations in the tapetum vary with trauma type, chronicity, treatment, and whether they are associated with specific symptom clusters. The current study excluded pediatric cases, so additional research on white matter disruption in pediatric trauma and PTSD is warranted. Lastly, while we considered comorbidities as potential confounding variables, we did not examine their association with dMRI metrics. Future collaborations with the ENIGMA Brain Injury, MDD, and Addiction working groups will provide opportunities to separate general neuroimaging biomarkers of psychopathology and disorder-specific effects.

## Conclusions

Here we presented results from the PGC-ENIGMA PTSD working group, reporting poorer white matter organization in the tapetum in individuals currently suffering from PTSD. We present the largest dMRI study in PTSD to date and the first to use harmonized image processing across sites, increasing our power to detect subtle effects. While future studies need to confirm the involvement of the tapetum specifically, our results add to the existing literature implicating the hippocampus and associated white matter connections as neural markers of structural disruptions in PTSD.

### Code availability

All analyses were conducted using generalizable scripts available on the ENIGMA-GitHub https://github.com/ENIGMA-git/ENIGMA/tree/master/WorkingGroups/EffectSize_and_GLM. Individual ROI level data were shared with the central site and processed using a set of R scripts with regressions customized for the current PGC-ENIGMA-PTSD dMRI analysis, publicly available on a set of Google Spreadsheet configuration files.

**Acknowledgements** K99NS096116 to ELD; CIHR, CIMVHR; Dana Foundation (to JBN); the University of Wisconsin Institute for Clinical and Translational Research (to Dr Emma Seppala); a National Science

Foundation Graduate Research Fellowship (to DWG); R01MH043454 and T32MH018931 (to RJD); and a core grant to the Waisman Center from the National Institute of Child Health and Human Development (P30HD003352); Defense and Veterans Brain Injury Centers, the U.S. Army Medical Research and Materiel Command (USAMRMC; W81XWH-13-2-0025) and the Chronic Effects of Neurotrauma Consortium (CENC; PT108802-SC104835); Department of Defense award number W81XWH-12-2-0012; ENIGMA was also supported in part by NIH U54EB020403 from the Big Data to Knowledge (BD2K) program, R56AG058854, R01MH116147, R01MH111671, and P41EB015922; Department of Veterans Affairs via support for the National Center for PTSD, NIAAA via its support for (P50) Center for the Translational Neuroscience if Alcohol, and NCATS via its support of (CTSA) Yale Center for Clinical Investigation; DoD W81XWH-10-1-0925; Center for Brain and Behavior Research Pilot Grant; South Dakota Governor's Research Center Grant; F32MH109274; Funding from the Bill & Melinda Gates Foundation; Funding from the SAMRC Unit on Risk & Resilience in Mental Disorders; German Research Foundation grant to JKD (numbers DA 1222/4-1 and WA 1539/8-2).; German Research Society (Deutsche Forschungsgemeinschaft, DFG; SFB/TRR 58: C06, C07); I01-CX000715 & I01-CX001542; K01MH118428; K23MH090366; K2CX001772, Clinical Science Research and Development Service, VA Office of Research and Development; MH098212; MH071537; M01RR00039; UL1TR000454; HD071982; HD085850; MH101380; NARSAD 27040; NARSAD Young Investigator, K01MH109836, Young Investigator Grant, Korean Scientists and Engineers Association; NHMRC Program Grant # 1073041; R01AG050595, R01AG022381; R01AG058822; R01EB015611; R01MH105355; R01MH105355, R01DA035484; R01MH111671, R01MH117601, R01AG059874, MJFF 14848; R01MH111671; VISN6 MIRECC; R21MH112956, Anonymous Women's Health Fund, Kasparian Fund, Trauma Scholars Fund, Barlow Family Fund; South African Medical Research Council for the "Shared Roots" Flagship Project, Grant no. MRC-RFA-IFSP-01-2013/SHARED ROOTS" through funding received from the South African National Treasury under its Economic Competitiveness and Support Package. The work by LLvdH reported herein was made possible through funding by the South African Medical Research Council through its Division of Research Capacity Development under the SAMRC CLINICIAN RESEARCHER (M.D PHD) SCHOLARSHIP PROGRAMME from funding received from the South African National Treasury.; South African Research Chairs Initiative in Posttraumatic Stress Disorder through the Department of Science and Technology and the National Research Foundation.; the National Natural Science Foundation of China (No. 31271099, 31471004), the Key Research Program of the Chinese Academy of Sciences (No. ZDRW-XH-2019-4); The study was supported by ZonMw, the Netherlands organization for Health Research and Development (40-00812-98-10041), and by a grant from the Academic Medical Center Research Council (110614) both awarded to MO.; Translational Research Center for TBI and Stress Disorders (TRACTS), a VA Rehabilitation Research and Development (RR&D) Traumatic Brain Injury Center of Excellence (B9254-C) at VA Boston Healthcare System.; VA CSR&D 1IK2CX001680; VISN17 Center of Excellence Pilot funding; VA CSR&D 822-MR-18176 and Senior Career Scientist Award; VA National Center for PTSD; VA RR&D 1IK2RX000709; VA RR&D 1K1RX002325; 1K2RX002922; VA RR&D I01RX000622; CDMRP W81XWH-08–2–0038; W81XWH08-2-0159 (PI: MBS) from the US Department of Defense. The views reflected here are strictly those of the authors and do not constitute endorsement by any of the funding sources listed here.

**Author contributions** ELD, SGD, NF, LES, ML, EKC, NJ, and RAM wrote the initial draft, all authors reviewed and edited the manuscript. All authors contributed data to the analyses.

## Compliance with ethical standards

**Conflict of interest** CGA has served as a consultant, speaker and/or on advisory boards for FSV7, Lundbeck, Genentech and Janssen, and editor of Chronic Stress for Sage Publications, Inc.; he has filed a patent for using mTOR inhibitors to augment the effects of antidepressants (filed on August 20, 2018). RJD is the founder and president of, and serves on the board of directors for, the nonprofit organization Healthy Minds Innovations, Inc. JK is a consultant for AbbVie, Inc., Amgen, Astellas Pharma Global Development, Inc., AstraZeneca Pharmaceuticals, Biomedisyn Corporation, Bristol-Myers Squibb, Eli Lilly and Company, Euthymics Bioscience, Inc., Neurovance, Inc., FORUM Pharmaceuticals, Janssen Research & Development, Lundbeck Research USA, Novartis Pharma AG, Otsuka America Pharmaceutical, Inc., Sage Therapeutics, Inc., Sunovion Pharmaceuticals, Inc., and Takeda Industries; is on the Scientific Advisory Board for Lohocla Research Corporation, Mnemosyne Pharmaceuticals, Inc., Naurex, Inc., and Pfizer; is a stockholder in Biohaven Pharmaceuticals; holds stock options in Mnemosyne Pharmaceuticals, Inc.; holds patents for Dopamine and Noradrenergic Reuptake Inhibitors in Treatment of Schizophrenia, US Patent No. 5,447,948 (issued September 5, 1995), and Glutamate Modulating Agents in the Treatment of Mental Disorders, U.S. Patent No. 8,778,979 (issued July 15, 2014); and filed a patent for Intranasal Administration of Ketamine to Treat Depression. U.S. Application No. 14/197,767 (filed on March 5, 2014); US application or Patent Cooperation Treaty international application No. 14/306,382 (filed on June 17, 2014). Filed a patent for using mTOR inhibitors to augment the effects of antidepressants (filed on August 20, 2018). NJ received partial research support from Biogen, Inc. (Boston, USA) for research unrelated to the content of this paper. PMT received partial research support from Biogen, Inc. (Boston, USA) for research unrelated to the topic of this paper.

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

## References

1. Atwoli L, Stein DJ, Koenen KC, McLaughlin KA. Epidemiology of posttraumatic stress disorder: prevalence, correlates and consequences. Curr Opin Psychiatry. 2015;28:307–11.
2. Fulton JJ, Calhoun PS, Wagner HR, Schry AR, Hair LP, Feeling N, et al. The prevalence of posttraumatic stress disorder in Operation Enduring Freedom/Operation Iraqi Freedom (OEF/OIF) Veterans: a meta-analysis. J Anxiety Disord. 2015;31:98–107.
3. Ferry F, Bunting B, Murphy S, O'Neill S, Stein D, Koenen K. Traumatic events and their relative PTSD burden in Northern Ireland: a consideration of the impact of the 'Troubles'. Soc Psychiatry Psychiatr Epidemiol. 2014;49:435–46.
4. Galea S, Nandi A, Vlahov D. The epidemiology of post-traumatic stress disorder after disasters. Epidemiol Rev. 2005;27:78–91.
5. Uno H, Eisele S, Sakai A, Shelton S, Baker E, DeJesus O, et al. Neurotoxicity of glucocorticoids in the primate brain. Horm Behav. 1994;28:336–48.
6. Antonow-Schlorke I, Helgert A, Gey C, Coksaygan T, Schubert H, Nathanielsz PW, et al. Adverse effects of antenatal glucocorticoids on cerebral myelination in sheep. Obstet Gynecol. 2009;113:142–51.
7. Basser PJ, Mattiello J, LeBihan D. MR diffusion tensor spectroscopy and imaging. Biophys J. 1994;66:259–67.
8. Bolzenius JD, Velez CS, Lewis JD, Bigler ED, Wade BSC, Cooper DB, et al. Diffusion imaging findings in US service members with mild traumatic brain injury and posttraumatic stress disorder. J Head Trauma Rehabil. 2018;33:393–402.

9. Durkee CA, Sarlls JE, Hommer DW, Momenan R. White matter microstructure alterations: a study of alcoholics with and without post-traumatic stress disorder. PLoS ONE. 2013;8:e80952.
10. Fani N, King TZ, Jovanovic T, Glover EM, Bradley B, Choi K, et al. White matter integrity in highly traumatized adults with and without post-traumatic stress disorder. Neuropsychopharmacology. 2012;37:2740–6.
11. Fani N, King TZ, Shin J, Srivastava A, Brewster RC, Jovanovic T, et al. Structural and functional connectivity in posttraumatic stress disorder: associations with FKBP5. Depress Anxiety. 2016;33:300–7.
12. Kim SJ, Jeong D-U, Sim ME, Bae SC, Chung A, Kim MJ, et al. Asymmetrically altered integrity of cingulum bundle in posttraumatic stress disorder. Neuropsychobiology. 2006;54:120–5.
13. Kim MJ, Lyoo IK, Kim SJ, Sim M, Kim N, Choi N, et al. Disrupted white matter tract integrity of anterior cingulate in trauma survivors. Neuroreport. 2005;16:1049–53.
14. Koch SBJ, van Zuiden M, Nawijn L, Frijling JL, Veltman DJ, Olff M. Decreased uncinate fasciculus tract integrity in male and female patients with PTSD: a diffusion tensor imaging study. J Psychiatry Neurosci. 2017;42:331–42.
15. Lepage C, de Pierrefeu A, Koerte IK, Coleman MJ, Pasternak O, Grant G, et al. White matter abnormalities in mild traumatic brain injury with and without post-traumatic stress disorder: a subject-specific diffusion tensor imaging study. Brain Imaging Behav. 2018;12:870–81.
16. O'Doherty DCM, Ryder W, Paquola C, Tickell A, Chan C, Hermens DF, et al. White matter integrity alterations in post-traumatic stress disorder. Hum Brain Mapp. 2018; 39:1327–38.
17. Olson EA, Cui J, Fukunaga R, Nickerson LD, Rauch SL, Rosso IM. Disruption of white matter structural integrity and connectivity in posttraumatic stress disorder: a TBSS and tractography study. Depress Anxiety. 2017;34:437–45.
18. Sanjuan PM, Thoma R, Claus ED, Mays N, Caprihan A. Reduced white matter integrity in the cingulum and anterior corona radiata in posttraumatic stress disorder in male combat veterans: a diffusion tensor imaging study. Psychiatry Res. 2013;214:260–8.
19. Santhanam P, Teslovich T, Wilson SH, Yeh P-H, Oakes TR, Weaver LK. Decreases in white matter integrity of ventro-limbic pathway linked to post-traumatic stress disorder in mild traumatic brain injury. J Neurotrauma. 2019;36:1093–8.
20. Schuff N, Zhang Y, Zhan W, Lenoci M, Ching C, Boreta L, et al. Patterns of altered cortical perfusion and diminished subcortical integrity in posttraumatic stress disorder: an MRI study. Neuroimage. 2011;54:S62–8.
21. Sun Y, Wang Z, Ding W, Wan J, Zhuang Z, Zhang Y, et al. Alterations in white matter microstructure as vulnerability factors and acquired signs of traffic accident-induced PTSD. PLoS ONE. 2013;8:e83473.
22. Sun Y-W, Hu H, Wang Y, Ding W-N, Chen X, Wan J-Q, et al. Inter-hemispheric functional and anatomical connectivity abnormalities in traffic accident-induced PTSD: a study combining fMRI and DTI. J Affect Disord. 2015;188:80–88.
23. Wang H-H, Zhang Z-J, Tan Q-R, Yin H, Chen Y-C, Wang H-N, et al. Psychopathological, biological, and neuroimaging characterization of posttraumatic stress disorder in survivors of a severe coalmining disaster in China. J Psychiatr Res. 2010;44:385–92.
24. Hu H, Zhou Y, Wang Q, Su S, Qiu Y, Ge J, et al. Association of abnormal white matter integrity in the acute phase of motor vehicle accidents with post-traumatic stress disorder. J Affect Disord. 2016;190:714–22.
25. Abe O, Yamasue H, Kasai K, Yamada H, Aoki S, Iwanami A, et al. Voxel-based diffusion tensor analysis reveals aberrant anterior cingulum integrity in posttraumatic stress disorder due to terrorism. Psychiatry Res. 2006;146:231–42.
26. Aschbacher K, Mellon SH, Wolkowitz OM, Henn-Haase C, Yehuda R, Flory JD, et al. Posttraumatic stress disorder, symptoms, and white matter abnormalities among combat-exposed veterans. Brain Imaging Behav. 2018;12:989–99.
27. Averill CL, Averill LA, Wrocklage KM, Scott JC, Akiki TJ, Schweinsburg B, et al. Altered white matter diffusivity of the cingulum angular bundle in posttraumatic stress disorder. Mol Neuropsychiatry. 2018;4:75–82.
28. Bierer LM, Ivanov I, Carpenter DM, Wong EW, Golier JA, Tang CY, et al. White matter abnormalities in Gulf War veterans with posttraumatic stress disorder: a pilot study. Psychoneuroendocrinology. 2015;51:567–76.
29. Davenport ND, Lim KO, Sponheim SR. White matter abnormalities associated with military PTSD in the context of blast TBI. Hum Brain Mapp. 2015;36:1053–64.
30. Li L, Lei D, Li L, Huang X, Suo X, Xiao F, et al. White matter abnormalities in post-traumatic stress disorder following a specific traumatic event. EBioMedicine. 2016;4:176–83.
31. Weis CN, Belleau EL, Pedersen WS, Miskovich TA, Larson CL. Structural connectivity of the posterior cingulum is related to reexperiencing symptoms in posttraumatic stress disorder. Chronic Stress. 2018;2. https://doi.org/10.1177/2470547018807134.
32. Zhang L, Zhang Y, Li L, Li Z, Li W, Ma N, et al. Different white matter abnormalities between the first-episode, treatment-naive patients with posttraumatic stress disorder and generalized anxiety disorder without comorbid conditions. J Affect Disord. 2011;133:294–9.
33. Dretsch MN, Lange RT, Katz JS, Goodman A, Daniel TA, Deshpande G, et al. Examining microstructural white matter in active duty soldiers with a history of mild traumatic brain injury and traumatic stress. Open Neuroimag J. 2017;11:46–57.
34. Maksimovskiy AL, McGlinchey RE, Fortier CB, Salat DH, Milberg WP, Oscar-Berman M. White matter and cognitive changes in veterans diagnosed with alcoholism and PTSD. J Alcohol Drug Depend. 2014;2:144.
35. Morey RA, Haswell CC, Selgrade ES, Massoglia D, Liu C, Weiner J, et al. Effects of chronic mild traumatic brain injury on white matter integrity in Iraq and Afghanistan war veterans. Hum Brain Mapp. 2013;34:2986–99.
36. Kennis M, van Rooij SJH, van den Heuvel MP, Kahn RS, Geuze E. Functional network topology associated with posttraumatic stress disorder in veterans. Neuroimage Clin. 2016;10:302–9.
37. Logue MW, van Rooij SJH, Dennis EL, Davis SL, Hayes JP, Stevens JS et al. Smaller hippocampal volume in posttraumatic stress disorder: a multi-site ENIGMA-PGC study. Biol Psychiatry 2018;83:244–53.
38. Jahanshad N, Kochunov PV, Sprooten E, Mandl RC, Nichols TE, Almasy L, et al. Multi-site genetic analysis of diffusion images and voxelwise heritability analysis: a pilot project of the ENIGMA–DTI working group. Neuroimage. 2013;81:455–69.
39. Kelly S, Jahanshad N, Zalesky A, Kochunov P, Agartz I, Alloza C, et al. Widespread white matter microstructural differences in schizophrenia across 4322 individuals: results from the ENIGMA Schizophrenia DTI Working Group. Mol Psychiatry. 2018;23:1261–9.
40. Favre P, Pauling M, Stout J, Hozer F, Sarrazin S, Abé C, et al. ENIGMA Bipolar Disorder Working Group. Widespread white matter microstructural abnormalities in bipolar disorder: evidence from mega- and meta-analyses across 3033 individuals. Neuropsychopharmacology. 2019:1–9.
41. Kelly S, van Velzen L, Veltman D, Thompson P, Jahanshad N, Schmaal L, et al. 941. White Matter Microstructural Differences in Major Depression: Meta-Analytic Findings from Enigma-MDD DTI. Biol Psychiatry. 2017;81:S381.

42. Villalon-Reina JE, Ching CRK, Kothapalli D, Sun D, Nir T, Lin A et al. Alternative diffusion anisotropy measures for the investigation of white matter alterations in 22q11.2 deletion syndrome. In: 14th International Symposium on Medical Information Processing and Analysis. International Society for Optics and Photonics, 2018, p 109750U.
43. Piras F, Piras F, Abe Y, Agarwal SM, Anticevic A, Ameis S, et al. White matter microstructure and its relation to clinical features of obsessive-compulsive disorder: findings from the ENIGMA OCD Working Group. BioRxiv. 2019:855916.
44. Asmundson GJ, Frombach I, McQuaid J, Pedrelli P, Lenox R, Stein MB. Dimensionality of posttraumatic stress symptoms: a confirmatory factor analysis of DSM-IV symptom clusters and other symptom models. Behav Res Ther. 2000;38:203–14.
45. Nyholt DR. A simple correction for multiple testing for single-nucleotide polymorphisms in linkage disequilibrium with each other. Am J Hum Genet. 2004;74:765–9.
46. Li J, Ji L. Adjusting multiple testing in multilocus analyses using the eigenvalues of a correlation matrix. Heredity. 2005;95:221–7.
47. Hendricks AE, Dupuis J, Logue MW, Myers RH, Lunetta KL. Correction for multiple testing in a gene region. Eur J Hum Genet. 2014;22:414–8.
48. Kochunov P, Williamson DE, Lancaster J, Fox P, Cornell J, Blangero J et al. Fractional anisotropy of water diffusion in cerebral white matter across the lifespan. Neurobiol Aging 2010;33:1–12.
49. De Bellis MD, Hooper SR, Sapia JL. Early trauma exposure and the brain. In: Vasterling JJ (ed). Neuropsychology of PTSD: Biological, cognitive, and clinical perspectives. New York, NY, US: The Guilford Press; 2005, pp 153–77.
50. Geibprasert S, Gallucci M, Krings T. Alcohol-induced changes in the brain as assessed by MRI and CT. Eur Radio. 2010;20:1492–501.
51. Pechtel P, Pizzagalli DA. Effects of early life stress on cognitive and affective function: an integrated review of human literature. Psychopharmacology. 2011;214:55–70.
52. Grant BF, Dawson DA, Stinson FS, Chou SP, Dufour MC, Pickering RP. The 12-month prevalence and trends in DSM-IV alcohol abuse and dependence: United States, 1991–1992 and 2001–2002. Drug Alcohol Depend. 2004;74:223–34.
53. Jovanovic T, Norrholm SD, Fennell JE, Keyes M, Fiallos AM, Myers KM, et al. Posttraumatic stress disorder may be associated with impaired fear inhibition: relation to symptom severity. Psychiatry Res. 2009;167:151–60.
54. Milad MR, Pitman RK, Ellis CB, Gold AL, Shin LM, Lasko NB, et al. Neurobiological basis of failure to recall extinction memory in posttraumatic stress disorder. Biol Psychiatry. 2009;66:1075–82.
55. Maller JJ, Welton T, Middione M, Callaghan FM, Rosenfeld JV, Grieve SM. Revealing the Hippocampal Connectome through Super-Resolution 1150-Direction Diffusion MRI. Sci Rep. 2019;9. https://doi.org/10.1038/s41598-018-37905-9.
56. Simmonds DJ, Hallquist MN, Asato M, Luna B. Developmental stages and sex differences of white matter and behavioral development through adolescence: a longitudinal diffusion tensor imaging (DTI) study. NeuroImage 2014;92:356–68.
57. Jin C, Qi R, Yin Y, Hu X, Duan L, Xu Q, et al. Abnormalities in whole-brain functional connectivity observed in treatment-naive post-traumatic stress disorder patients following an earthquake. Psychol Med. 2014;44:1927–36.
58. Miller DR, Hayes SM, Hayes JP, Spielberg JM, Lafleche G, Verfaellie M. Default mode network subsystems are differentially disrupted in posttraumatic stress disorder. Biol Psychiatry Cogn Neurosci Neuroimaging. 2017;2:363–71.
59. Malivoire BL, Girard TA, Patel R, Monson CM. Functional connectivity of hippocampal subregions in PTSD: relations with symptoms. BMC Psychiatry. 2018;18:129.
60. Fani N, King TZ, Reiser E, Binder EB, Jovanovic T, Bradley B, et al. FKBP5 genotype and structural integrity of the posterior cingulum. Neuropsychopharmacology. 2014;39:1206–13.
61. Rohlfing T. Incorrect ICBM-DTI-81 atlas orientation and white matter labels. Front Neurosci. 2013;7:4.
62. Lyon M, Welton T, Varda A, Maller JJ, Broadhouse K, Korgaonkar MS, et al. Gender-specific structural abnormalities in major depressive disorder revealed by fixel-based analysis. Neuroimage Clin. 2019;21:101668.
63. Chen L, Hu X, Ouyang L, He N, Liao Y, Liu Q, et al. A systematic review and meta-analysis of tract-based spatial statistics studies regarding attention-deficit/hyperactivity disorder. Neurosci Biobehav Rev. 2016;68:838–47.
64. Cloitre M, Stolbach BC, Herman JL, van der Kolk B, Pynoos R, Wang J, et al. A developmental approach to complex PTSD: childhood and adult cumulative trauma as predictors of symptom complexity. J Trauma Stress. 2009;22:399–408.
65. Hart H, Rubia K. Neuroimaging of child abuse: a critical review. Front Hum Neurosci. 2012;6:52.
66. Campbell DG, Felker BL, Liu C-F, Yano EM, Kirchner JE, Chan D, et al. Prevalence of depression–PTSD comorbidity: implications for clinical practice guidelines and primary care-based interventions. J Gen Intern Med. 2007;22:711–8.
67. Sexton CE, Mackay CE, Ebmeier KP. A systematic review of diffusion tensor imaging studies in affective disorders. Biol Psychiatry. 2009;66:814–23.
68. Shen X, Reus LM, Cox SR, Adams MJ, Liewald DC, Bastin ME, et al. Subcortical volume and white matter integrity abnormalities in major depressive disorder: findings from UK Biobank imaging data. Sci Rep. 2017;7:5547.
69. van Velzen LS, Kelly S, Isaev D, Aleman A, Aftanas LI, Bauer J et al. White matter disturbances in major depressive disorder: a coordinated analysis across 20 international cohorts in the ENIGMA MDD working group. Mol Psychiatry. 2019. https://doi.org/10.1038/s41380-019-0477-2.
70. Rogers JM, Read CA. Psychiatric comorbidity following traumatic brain injury. Brain Inj. 2007;21:1321–33.
71. Johnson VE, Stewart W, Smith DH. Axonal pathology in traumatic brain injury. Exp Neurol. 2013;246:35–43.
72. Dennis EL, Wilde EA, Newsome MR, Scheibel RS, Troyanskaya M, Velez C, et al. Enigma military brain injury: a coordinated meta-analysis of diffusion MRI from multiple cohorts. Proc IEEE Int Symp Biomed Imaging. 2018;2018:1386–9.
73. Hunsberger J, Austin DR, Henter ID, Chen G. The neurotrophic and neuroprotective effects of psychotropic agents. Dialogues Clin Neurosci. 2009;11:333.
74. Brown PJ, Stout RL, Mueller T. Posttraumatic stress disorder and substance abuse relapse among women: a pilot study. Psychol Addict Behav. 1996;10:124–8.
75. Brown PJ, Stout RL, Mueller T. Substance use disorder and posttraumatic stress disorder comorbidity: Addiction and psychiatric treatment rates. Psychol Addict Behav; 1999;13:115–22.
76. Cardenas VA, Greenstein D, Fouche J-P, Ferrett H, Cuzen N, Stein DJ, et al. Not lesser but greater fractional anisotropy in adolescents with alcohol use disorders. Neuroimage Clin. 2013;2:804–9.
77. Tapert SF, Theilmann RJ, Schweinsburg AD, Yafai S, Frank LR. Reduced fractional anisotropy in the splenium of adolescents with alcohol use disorder. Age 2003;16. https://cds.ismrm.org/ismrm-2003/2241.pdf.
78. Thompson P, Jahanshad N, Ching CRK, Salminen L, Thomopoulos SI, Bright J et al. ENIGMA and global neuroscience: a

decade of large-scale studies of the brain in health and disease across more than 40 countries. PsyArXiv. 2019.
79. Gilbertson MW, Shenton ME, Ciszewski A, Kasai K, Lasko NB, Orr SP, et al. Smaller hippocampal volume predicts pathologic vulnerability to psychological trauma. Nat Neurosci. 2002;5:1242–7.
80. Admon R, Leykin D, Lubin G, Engert V, Andrews J, Pruessner J, et al. Stress-induced reduction in hippocampal volume and connectivity with the ventromedial prefrontal cortex are related to maladaptive responses to stressful military service. Hum Brain Mapp. 2013;34:2808–16.
81. Galinowski A, Miranda R, Lemaitre H, Paillère Martinot M-L, Artiges E, Vulser H, et al. Resilience and corpus callosum microstructure in adolescence. Psychol Med. 2015;45:2285–94.
82. Vlasova RM, Siddarth P, Krause B, Leaver AM, Laird KTSt, Cyr N, et al. Resilience and white matter integrity in geriatric depression. Am J Geriatr Psychiatry. 2018;26:874–83.
83. Taliaz D, Loya A, Gersner R, Haramati S, Chen A, Zangen A. Resilience to chronic stress is mediated by hippocampal brain-derived neurotrophic factor. J Neurosci. 2011;31:4475–83.
84. Krishnan V, Han M-H, Graham DL, Berton O, Renthal W, Russo SJ, et al. Molecular adaptations underlying susceptibility and resistance to social defeat in brain reward regions. Cell. 2007;131:391–404.
85. Frodl T, Schüle C, Schmitt G, Born C, Baghai T, Zill P, et al. Association of the brain-derived neurotrophic factor Val66Met polymorphism with reduced hippocampal volumes in major depression. Arch Gen Psychiatry. 2007;64:410.
86. Bathina S, Das UN. Brain-derived neurotrophic factor and its clinical implications. Arch Med Sci. 2015;11:1164–78.
87. Frodl T, Carballedo A, Fagan AJ, Lisiecka D, Ferguson Y, Meaney JF. Effects of early-life adversity on white matter diffusivity changes in patients at risk for major depression. J Psychiatry Neurosci. 2012;37:37–45.
88. Kochunov P, Jahanshad N, Sprooten E, Nichols TE, Mandl RC, Almasy L, et al. Multi-site study of additive genetic effects on fractional anisotropy of cerebral white matter: comparing meta and megaanalytical approaches for data pooling. Neuroimage. 2014;95:136–50.
89. Boedhoe PS, Schmaal L, Abe Y, Ameis SH, Arnold PD, Batistuzzo MC, et al. Distinct subcortical volume alterations in pediatric and adult OCD: a worldwide meta- and mega-analysis. Am J Psychiatry. 2017;174:60–69.

## Affiliations

**Emily L. Dennis[1,2,3,4] · Seth G. Disner[5,6] · Negar Fani[7] · Lauren E. Salminen[2] · Mark Logue[8,9,10,11] · Emily K. Clarke[12,13] · Courtney C. Haswell[12,13] · Christopher L. Averill[14] · Lee A. Baugh[15,16,17] · Jessica Bomyea[18,19] · Steven E. Bruce[20] · Jiook Cha[21,22] · Kyle Choi[23] · Nicholas D. Davenport[5,6] · Maria Densmore[24,25] · Stefan du Plessis[26] · Gina L. Forster[15,16,27] · Jessie L. Frijling[28] · Atilla Gonenc[29,30] · Staci Gruber[29,30] · Daniel W. Grupe[31] · Jeffrey P. Guenette[32] · Jasmeet Hayes[33] · David Hofmann[34] · Jonathan Ipser[35] · Tanja Jovanovic[7,36] · Sinead Kelly[1,37] · Mitzy Kennis[38,39] · Philipp Kinzel[1,40] · Saskia B. J. Koch[28,41] · Inga Koerte[1,40] · Sheri Koopowitz[35] · Mayuresh Korgaonkar[42] · John Krystal[14] · Lauren A. M. Lebois[30,43] · Gen Li[44,45] · Vincent A. Magnotta[46] · Antje Manthey[47] · Geoff J. May[48,49,50,51] · Deleene S. Menefee[52,53] · Laura Nawijn[28,54] · Steven M. Nelson[48,49,51] · Richard W. J. Neufeld[24,55,56,57] · Jack B. Nitschke[58] · Daniel O'Doherty[59] · Matthew Peverill[60] · Kerry J. Ressler[7,30,41] · Annerine Roos[61] · Margaret A. Sheridan[62] · Anika Sierk[47] · Alan Simmons[18,19] · Raluca M. Simons[16,63] · Jeffrey S. Simons[17,63] · Jennifer Stevens[7] · Benjamin Suarez-Jimenez[21,22] · Danielle R. Sullivan[8,9] · Jean Théberge[24,25,64] · Jana K. Tran[65] · Leigh van den Heuvel[26] · Steven J. A. van der Werff[66,67] · Sanne J. H. van Rooij[7] · Mirjam van Zuiden[28] · Carmen Velez[3,68] · Mieke Verfaellie[9,69] · Robert R. J. M. Vermeiren[66] · Benjamin S. C. Wade[3,68,70] · Tor Wager[71] · Henrik Walter[47] · Sherry Winternitz[30,72] · Jonathan Wolff[43] · Gerald York[73,74] · Ye Zhu[44,45] · Xi Zhu[21,22] · Chadi G. Abdallah[14] · Richard Bryant[75] · Judith K Daniels[76] · Richard J Davidson[31,58,77] · Kelene A Fercho[15,16,17,78] · Carol Franz[19,79] · Elbert Geuze[38,39] · Evan M Gordon[48,49,51] · Milissa L Kaufman[30,72] · William S. Kremen[18,19,79] · Jim Lagopoulos[59] · Ruth A Lanius[24,25,56] · Michael J. Lyons[80] · Stephen R McCauley[81,82] · Regina McGlinchey[30,83] · Katie A. McLaughlin[84] · William Milberg[30,85] · Yuval Neria[21,22] · Miranda Olff[28,85] · Soraya Seedat[61] · Martha Shenton[1,86] · Scott R. Sponheim[5,6] · Dan J. Stein[35] · Murray B. Stein[19,87] · Thomas Straube[34] · David F. Tate[3,68] · Nic J. A. van der Wee[66,67] · Dick J. Veltman[54] · Li. Wang[44,45] · Elisabeth A. Wilde[3,82,88] · Paul M. Thompson[2] · Peter Kochunov[89] · Neda Jahanshad[2] · Rajendra A. Morey[12,13]**

[1] Psychiatry Neuroimaging Laboratory, Department of Psychiatry, Brigham and Women's Hospital, Boston, MA, USA

[2] Imaging Genetics Center, Stevens Neuroimaging & Informatics Institute, Keck School of Medicine of USC, Marina del Rey, CA, USA

[3] Department of Neurology, University of Utah, Salt Lake City, UT, USA

[4] Stanford Neurodevelopment, Affect, and Psychopathology Laboratory, Stanford, CA, USA

[5] Minneapolis VA Health Care System, Minneapolis, MN, USA

[6] Department of Psychiatry, University of Minnesota, Minneapolis, MN, USA

[7] Department of Psychiatry and Behavioral Sciences, Emory University School of Medicine, Atlanta, GA, USA

[8] National Center for PTSD, VA Boston Healthcare System, Boston, MA, USA

[9] Department of Psychiatry, Boston University School of Medicine, Boston, MA, USA

[10] Biomedical Genetics, Boston University School of Medicine, Boston, MA, USA

[11] Department of Biostatistics, Boston University School of Public Health, Boston, MA, USA

[12] Brain Imaging and Analysis Center, Duke University, Durham, NC, USA

[13] VISN 6 MIRECC, Durham VA, Durham, NC, USA

[14] Clinical Neuroscience Division, National Center for PTSD; Department of Psychiatry, Yale University School of Medicine, New Haven, CT, USA

[15] Division of Basic Biomedical Sciences, Sanford School of Medicine, University of South Dakota, Vermillion, SD, USA

[16] Center for Brain and Behavior Research, University of South Dakota, Vermillion, SD, USA

[17] Sioux Falls VA Health Care System, Sioux Falls, SD, USA

[18] Center of Excellence for Stress and Mental Health, VA San Diego Healthcare System, La Jolla, CA, USA

[19] Department of Psychiatry, University of California, San Diego, La Jolla, CA, USA

[20] Department of Psychological Sciences, Center for Trauma Recovery University of Missouri-St. Louis, St. Louis, MO, USA

[21] Department of Psychiatry, Columbia University Medical Center, New York, NY, USA

[22] New York State Psychiatric Institute, New York, NY, USA

[23] Health Services Research Center, University of California, San Diego, CA, USA

[24] Department of Psychiatry, Western University, London, ON, Canada

[25] Imaging Division, Lawson Health Research Institute, London, ON, Canada

[26] Department of Psychiatry, Stellenbosch University, Stellenbosch, South Africa

[27] Brain Health Research Centre, Department of Anatomy, University of Otago, Dunedin 9054, New Zealand

[28] Department of Psychiatry, Amsterdam Neuroscience, Amsterdam University Medical Centers, Location Academic Medical Center, University of Amsterdam, Amsterdam, The Netherlands

[29] Cognitive and Clinical Neuroimaging Core, McLean Hospital, Belmont, MA, USA

[30] Department of Psychiatry, Harvard Medical School, Boston, MA, USA

[31] Center for Healthy Minds, University of Wisconsin-Madison, Madison, WI, USA

[32] Division of Neuroradiology, Brigham and Women's Hospital, Boston, MA, USA

[33] Department of Psychology, The Ohio State University, Columbus, OH, USA

[34] Institute of Medical Psychology and Systems Neuroscience, University of Münster, Münster, Germany

[35] SA Medical Research Council Unit on Risk & Resilience in Mental Disorders, Dept of Psychiatry & Neuroscience Institute, University of Cape Town, Cape Town, South Africa

[36] Department of Psychiatry and Behavioral Neuroscience, Wayne State University School of Medicine, Detroit, MI, USA

[37] Department of Psychiatry, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA, USA

[38] Brain Center Rudolf Magnus, Department of Psychiatry, UMCU, Utrecht, The Netherlands

[39] Brain Research and Innovation Centre, Ministry of Defence, Utrecht, The Netherlands

[40] Department of Child and Adolescent Psychiatry, Psychosomatics and Psychotherapy, University Hospital, Ludwig-Maximilians-Universität, Munich, Germany

[41] Donders Institute for Brain, Cognition and Behavior, Centre for Cognitive Neuroimaging, Radboud University Nijmegen, Nijmegen, The Netherlands

[42] Brain Dynamics Centre, Westmead Institute of Medical Research, University of Sydney, Westmead, NSW, Australia

[43] Division of Depression and Anxiety Disorders, McLean Hospital, Belmont, MA, USA

[44] Laboratory for Traumatic Stress Studies, CAS Key Laboratory of Mental Health, Institute of Psychology, Chinese Academy of Sciences, Beijing, China

[45] Department of Psychology, University of Chinese Academy of Sciences, Beijing, China

[46] Departments of Radiology, Psychiatry, and Biomedical Engineering, University of Iowa, Iowa City, IA, USA

[47] University Medical Centre Charite, Berlin, Germany

[48] VISN 17 Center of Excellence for Research on Returning War Veterans, Waco, TX, USA

[49] Department of Psychology and Neuroscience, Baylor University, Waco, TX, USA

[50] Department of Psychiatry and Behavioral Science, Texas A&M Health Science Center, Bryan, TX, USA

[51] Center for Vital Longevity, School of Behavioral and Brain Sciences, University of Texas at Dallas, Dallas, TX, USA

[52] Menninger Department of Psychiatry, Baylor College of Medicine, Houston, TX, USA

[53] South Central MIRECC, Houston, TX, USA

[54] Department of Psychiatry, Amsterdam University Medical Centers, Location VU University Medical Center, VU University, Amsterdam, The Netherlands

[55] Department of Psychology, Western University, London, ON, Canada

[56] Department of Neuroscience, Western University, London, ON, Canada

[57] Department of Psychology, University of British Columbia, Okanagan, BC, Canada

[58] Department of Psychiatry, University of Wisconsin-Madison, Madison, WI, USA

[59] University of the Sunshine Coast, Birtinya, QLD, Australia

[60] Department of Psychology, University of Washington, Seattle, WA, USA

[61] South African Medical Research Council / Stellenbosch University Genomics of Brain Disorders Research Unit, Department of Psychiatry, Stellenbosch University, Cape Town, South Africa

[62] Department of Psychology and Brain Sciences, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA

[63] Department of Psychology, University of South Dakota, Vermillion, SD, USA

[64] Department of Medical Biophysics, Western University, London, ON, Canada

[65] Houston Fire Department, Houston, TX, USA

[66] Department of Psychiatry, LUMC, Leiden, The Netherlands

[67] Leiden Institute for Brain and Cognition, Leiden, The Netherlands

[68] Missouri Institute of Mental Health and University of Missouri, St Louis, MO, USA

[69] Memory Disorders Research Center, VA Boston Healthcare System, Boston, MA, USA

[70] Ahmanson-Lovelace Brain Mapping Center, Department of Neurology, University of California, Los Angeles, Los Angeles, CA, USA

[71] Dartmouth College, Hanover, NH, USA

[72] Division of Women's Mental Health, McLean Hospital, Belmont, MA, USA

[73] Joint Trauma System, 3698 Chambers Pass, Joint Base San Antonio, Fort Sam Houston, TX, USA

[74] Alaska Radiology Associates, Anchorage, AK, USA

[75] School of Psychology, University of New South Wales, Sydney, NSW, Australia

[76] Department of Clinical Psychology, University of Groningen, Groningen, The Netherlands

[77] Department of Psychology, University of Wisconsin-Madison, Madison, WI, USA

[78] Civil Aerospace Medical Institute, US Federal Aviation Administration, Oklahoma City, OK, USA

[79] Center for Behavior Genetics of Aging, University of California, San Diego, La Jolla, CA, USA

[80] Dept. of Psychological & Brain Sciences, Boston University, Boston, MA, USA

[81] Departments of Neurology and Pediatrics, Baylor College of Medicine, Houston, TX, USA

[82] H. Ben Taub Department of Physical Medicine and Rehabilitation, Baylor College of Medicine, Houston, TX, USA

[83] Geriatric Research Educational and Clinical Center and Translational Research Center for TBI and Stress Disorders, VA Boston Healthcare System, Boston, MA, USA

[84] Department of Psychology, Harvard University, Cambridge, MA, USA

[85] ARQ National Psychotrauma Centre, Diemen, The Netherlands

[86] VA Boston Healthcare System, Brockton Division, Brockton, MA, USA

[87] Department of Family Medicine and Public Health, University of California, San Diego, La Jolla, CA, USA

[88] George E. Whalen Veterans Affairs Medical Center, Salt Lake City, UT, USA

[89] Maryland Psychiatric Research Center, University of Maryland School of Medicine, Baltimore, MD, USA